Transporting Root Evidence for Repeated Adsorption Ranking

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***Abstract*—Scientific discovery pipelines increasingly execute repeated top-k ranking queries whose scoring function is neither a materialized attribute nor a closed-form algebraic expression, but an *implicit equilibrium score*—such as Ideal Adsorbed Solution Theory (IAST) mixture selectivity—defined by the numerical root of a query-parameterized nonlinear balance equation over stored piecewise-linear (PWL) curves. Existing top-k processors either recompute query-local bounds from scratch or reuse prior solutions solely to warm-start iterative root solvers *after* a candidate has been selected for probing, leaving the dominant cost of unpruned spreading-pressure integral evaluations untouched across query streams. We introduce Conditional Root-Enclosure Transport, an exact query-processing mechanism that elevates historical root-finding evidence from a post-filter solver warm start into a pre-probe candidate-pruning primitive. Exploiting opposite-monotone partial-pressure arguments on supported PWL isotherm segments, our method transports prior solved root brackets and one-sided negative threshold test witnesses into certified target-query root enclosures in $O(1)$ arithmetic per retained anchor, maps those enclosures into exact selectivity upper bounds, and couples them with a static support-cap index to prune non-competitive candidates before any target integral probe. Across 64-query streams over 16,010 `CRAFTED` and 353 `NIST--N`$_2$ adsorption pairs, `Transport` eliminates 78.3%–99.4% of query-phase spreading-pressure probes on `CRAFTED` and cuts end-to-end paid cold-pool lifecycle time (charging index construction, full seed acquisition, query execution, and teardown) to $0.46\times$–$0.63\times$ of query-local cap indexing (`LeanCap`), $0.50\times$–$0.74\times$ of incumbent continuation, and $0.43\times$–$0.54\times$ of solver warm-starting (`Seed--Warm`), while achieving $0.49\times$–$0.62\times$ against a seed-free `WarmOnDemand` baseline ($0.61\times$–$0.72\times$ at 1 bar). Diagnostic attribution shows that transported historical brackets drive 99.9% of all upper-bound exclusions on `CRAFTED` (949,984 vs. 704 static-cap exclusions in near $k = 1$), while delineating the exact operating boundaries where solved-root-only retention (`Roots--Only`) is 0%–8% faster and where unseeded baselines remain preferable on single-query prefixes, small pools with large k, wide random jumps, and 0.01 bar cold starts.**

***Index Terms*—scientific query processing, implicit scores, adsorption, repeated top-k queries, numerical reuse**

I. Introduction

Data-driven discovery engines increasingly execute repeated top-k ranking queries where a candidate tuple's score is neither a materialized attribute nor a closed-form vector similarity, but an *implicit equilibrium score* defined by the root of a query-parameterized nonlinear equation over stored functional curves [9], [10], [12]. A canonical instance is high-throughput porous materials screening under Ideal Adsorbed Solution Theory (IAST), which ranks candidate adsorbent pairs by their binary mixture selectivity across sweeping operating pressures P and feed compositions y [1], [2]. Across large repositories of experimental and simulated piecewise-linear (PWL) adsorption isotherms such as NIST ISODB and CRAFTED [3]–[5], the stored loading curves remain static, yet every arriving query $z = (P, y, k)$ shifts the nonlinear spreading-pressure balance equation—and thus the equilibrium root $x^* \in (0, 1)$ and selectivity ranking—for every candidate in the pool.

The central query-processing bottleneck in repeated implicit-score ranking is minimizing the number of expensive spreading-pressure integral probes $\Pi_i(p)$ required to certify the exact top-k result. Existing approaches address this bottleneck from two disconnected angles, each leaving the dominant source of work untouched (Fig. 1a). On the numerical side, domain libraries and equilibrium engines accelerate per-candidate solves via prefix integration, spline interpolation, and continuation warm starts [2], [6]–[8]. However, reusing a prior root merely as an initial guess inside a Newton or bisection solver (`Seed--Warm`) still enters the numerical kernel and evaluates target spreading-pressure integrals for every candidate that reaches the solver. On the database side, no-random-access (NRA) and minimal-probing top-k algorithms prune candidates by maintaining lower and upper score bounds [9]–[12]. While finite isotherm support yields a query-independent static cap index (`LeanCap`) that orders candidates by closed-form support-cap upper bounds, those query-local intervals remain wide ($w = 0.858$–0.893 on `CRAFTED`) and prune fewer than 0.1% of threshold tests once the active rank-k cut τ rises. Meanwhile, caching prior scalar selectivity scores cannot certify whether an unprobed candidate remains below the new rank-k boundary under a shifted query (P', y').

To bridge numerical continuation and bound-based top-k pruning, we propose **Conditional Root-Enclosure Transport**, an exact mechanism that transforms historical root-finding brackets into pre-probe selectivity upper bounds. Our key mathematical insight is that in binary IAST, the two pure-component fictitious pressures $p_1(x) = f_1/x = Py/x$ and $p_2(x) = f_2/(1-x) = P(1-y)/(1-x)$ vary strictly monotonically in *opposite* directions with respect to the adsorbed mole fraction $x \in (0, 1)$. Because spreading-pressure integrals $\Pi_i(p)$ are strictly increasing on supported domain segments, rescaling a source query's certified root enclosure $[\ell, u]$ by the directional partial-pressure ratios $c_1 = f_1'/f_1 = (P'y')/(Py)$

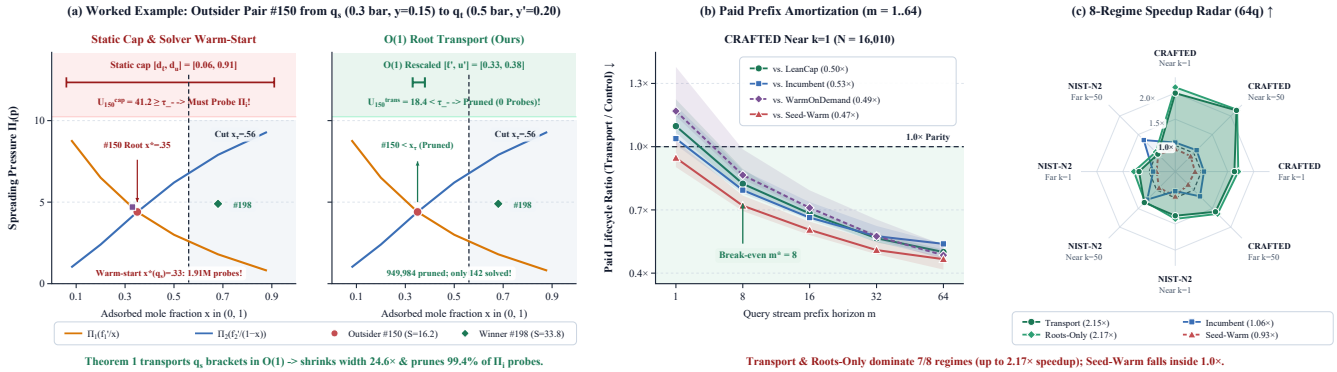


Fig. 1. **Static support caps and solver warm-starts still probe** 99.9% **of non-top-**k **candidates, whereas** $O(1)$ **root-enclosure transport prunes** 99.4% **of spreading-pressure probes before solver entry and cuts 64-query paid runtime by up to** 53%**. (a)** Query transition from $q_s(P = 0.3 \text{ bar}, y = 0.15)$ to $q_t(P' = 0.5 \text{ bar}, y' = 0.20)$ on non-top-k outsider pair `#150` ($S_{150} = 16.2 < \tau_- = 28.5$, cutoff mole fraction $x_\tau = 0.56$) vs. top-k winner `#198` ($S_{198} = 33.8 \geq \tau_-$): static support-cap intervals (`LeanCap`, $[d_\ell, d_u] = [0.06, 0.91]$) straddle x_τ ($U_{150}^{\text{cap}} = 41.2 \geq \tau_-$), while solver warm-starting (`Seed--Warm`) reuses $x^*(q_s) = 0.33$ only as an uncertified point guess inside Newton's solver, forcing 1.91M spreading-pressure probes $\Pi_i(p)$ (5.60 s). Our $O(1)$ directional transport (Theorem 1) rescales $[0.33, 0.33] \to [\ell', u'] = [0.33, 0.38]$ ($24.6\times$ narrower), certifying $U_{150}^{\text{trans}} = 18.4 < \tau_-$ and pruning 949,984 outsiders (99.4% of Π_i probes) with zero isotherm lookups (2.47–2.61 s). **(b)** Paid prefix amortization curves ($m \in \{1, 8, 16, 32, 64\}$, median and $[\min, \max]$ bands) breaking even by $m^* = 8$ queries. **(c)** 8-regime 64-query paid speedup radar profile relative to `LeanCap` ($1.0\times$ dashed parity ring), where `Transport` and `Roots--Only` dominate 7 of 8 regimes (up to $2.17\times$ speedup) while `Seed--Warm` falls inside the $1.0\times$ ring.

and $c_2 = f_2'/f_2 = (P'(1-y'))/(P(1-y))$ yields a certified target root enclosure $[\widehat{\ell}, \widehat{u}]$ in $O(1)$ floating-point operations without touching any target isotherm segment. Crucially, this transport law applies both to tightly solved equilibrium roots (candidates that survived a source threshold test) and to one-sided *negative threshold witnesses* $[\ell, u] = [d_\ell, x_\tau]$ recorded when a candidate fails a source threshold test $G_f(x_\tau) < 0$, shrinking mean target bracket widths by $24.6\times$ on `CRAFTED` near ($0.861 \to 0.035$) and $32.8\times$ on `NIST--N2` near ($0.970 \to 0.030$).

Turning transported root enclosures into a sound, cost-effective top-k query executor requires discharging three systems obligations. First, target root existence is *conditional*: a candidate that equilibrated at a source query (P, y) may exceed finite isotherm support at (P', y'), so transported bounds are valid only after passing an $O(1)$ linear support guard, and only currently eligible candidates with distinct pair IDs may advance the kth lower threshold τ. Second, retained brackets must be numerically certified: finite-precision rounding during a direct-score residual probe $H_f(\tau_-)$ must never fabricate a false sign change, which we enforce through outward-rounded 'long double' interval arithmetic, an explicit tolerance shell τ_-, and a 112-bit 'binary128' fallback solver. Third, the end-to-end cost of acquiring, storing, and transporting anchors must remain smaller than the integral probes they eliminate. We satisfy these obligations by coupling a fixed two-slot per-candidate anchor state (*first* + *recent*) with a static support-cap index and evaluating all policies under a strict *paid lifecycle* contract that charges constructor setup, index sorting, declared upfront seed acquisition, query execution, and teardown.

Our principal contributions are summarized as follows:

- **Exact Root-Enclosure Transport Theory (Sections III–IV):** We formalize repeated implicit-score top-k ranking over PWL isotherm relations and prove an $O(1)$ directional root-transport theorem (Theorem 1) with closed-form selectivity bounds and interval-width contraction guarantees (Proposition 1).
- **Guarded Cap-Indexed Ranking Executor (Section V):** We design a shared-kernel ranking executor that unifies static support-cap tail cutting, two-slot (*first* + *recent*) root and witness transport, and outward-rounded interval residual testing across 11 matched policies with zero output mismatches against canonical solving.
- **Paid Lifecycle Attribution and Crossover Analysis (Sections VI–VII):** Across 64-query streams on 16,010 `CRAFTED` and 353 `NIST--N2` pairs, we demonstrate that transported historical brackets drive 99.9% of upper-bound exclusions on `CRAFTED`, eliminating 78.3%–99.4% of query-phase $\Pi_i(p)$ probes and cutting paid cold-pool lifecycle time to $0.46\times$–$0.63\times$ of `LeanCap`, $0.50\times$–$0.74\times$ of `Incumbent`, and $0.43\times$–$0.54\times$ of `Seed--Warm` after breaking even by 8 near or 16 far queries.
- **WarmOnDemand and Operating-Boundary Characterization (Sections VII–VIII):** We benchmark against a seed-free `WarmOnDemand` baseline ($0.49\times$–$0.62\times$ on `CRAFTED` near/far and $0.61\times$–$0.72\times$ at 1 bar) and map the exact operating boundaries across pool sizes $N \in [256, 16{,}010]$, depths $k \in [1, 200]$, random arrival variances, and literature pressures (0.01–1 bar), identifying where solved-root-only retention (`Roots--Only`) is 0%–8% faster and where unseeded execution is preferable.

II. Related Work

IAST Equilibrium Solvers and Reusable Numerical State. Ideal Adsorbed Solution Theory (IAST) predicts multicomponent mixture adsorption equilibrium from single-component adsorption isotherms through spreading-pressure equalization [1]. Because the spreading-pressure balance has

no closed-form solution for general isotherms, domain packages such as `pyIAST` [2] and high-throughput screening routines [6] rely on analytical prefix integration over piecewise-linear (PWL) or analytical isotherm models coupled with Newton–Raphson or Brent root-finding. Recent simulation frameworks including `RUPTURA` [7] and restructured equilibrium solvers [8] accelerate repeated evaluations by caching curve prefixes and reusing prior equilibrium solutions as continuation warm starts. Our matched controls (`Incumbent` and `Seed--Warm`) incorporate these exact warm-start optimizations inside a shared numerical kernel. However, solver warm-starting only reduces iteration counts *after* a candidate enters the numerical solver; our work instead elevates historical root brackets into certified rank upper bounds that prune non-competitive candidates *before* any target residual probe occurs.

Top-k Query Processing with Expensive and Implicit Scores. Database top-k processing over expensive predicates builds on minimal-probing principles [9] and Fagin et al.'s No-Random-Access (NRA) paradigm [10], which maintain candidate lower and upper score bounds and terminate as soon as the kth lower bound exceeds all remaining upper bounds. Parameterized and query-dependent ranking extensions adapt bound maintenance when scoring weights vary across queries [11], while recent opaque-score retrieval allocates expensive probes statistically to approximate top-k answers [12]. Unlike linear or metric-space top-k retrieval where score bounds follow from triangle inequalities or dot-product envelopes, implicit equilibrium scores require bounding the uncomputed root of a query-dependent nonlinear equation. We bridge this gap by deriving closed-form root-enclosure transport rules from the opposite monotonicity of binary partial pressures, allowing exact NRA-style kth-lower-bound pruning to operate over both solved root brackets and one-sided negative threshold witnesses.

Certified Bound Reuse, Deferred Evaluation, and Scientific Benchmarks. Reusing certified intervals across related evaluations connects our approach to predicate range caching in `Casper` [13], safe screening rules and observed-value envelopes in convex optimization [14], [15], and lazy exact geometric predicates that refine interval approximations only on demand [16]. We specialize deferred interval refinement to monotone spreading-pressure curves, separating an exact-real directional containment proof (Theorem 1) from a floating-point guarded execution contract (δ-guarded sign certification and canonical fallback) over experimental isotherms from NIST ISODB [3], [5] and simulated metal-organic framework (MOF) isotherms from CRAFTED [4].

III. Finite-Curve Ranking Queries

We consider a scientific database relation $\mathcal{R}$ storing N candidate adsorption pairs, where each candidate $i \in \{1, \dots, N\}$ pairs two compatible pure-component isotherms (q_{i1}, q_{i2}) measured or simulated on the same adsorbent material at a fixed temperature and unit convention. A repeated ranking workload issues a stream of top-k queries $z = (P, y, k)$, where $P > 0$ denotes total mixture pressure, $y \in (0, 1)$ denotes the bulk mole fraction of component 1, and $k \geq 1$ is the requested ranking size. Defining the pure-component partial-pressure scales $f_1 = Py$ and $f_2 = P(1 - y)$, the query processor must return the k supported candidates with the highest binary mixture selectivity at (f_1, f_2).

A. Stored Curves and Supported Equilibria

Each stored pure-component isotherm $q_i(p)$ consists of positive loading knots at strictly increasing pressures $0 < p_{i,1} < \cdots < p_{i,m_i} = M_i$, interpolated piecewise linearly (PWL) between knots, connected linearly from the origin $(0, 0)$ to $(p_{i,1}, q_i(p_{i,1}))$, and undefined above its finite support limit M_i. While loading $q_i(p)$ need not be monotone across all pressures, positive loading ($q_i(p) > 0$ for $p \in (0, M_i]$) guarantees that the spreading-pressure integral

$$\Pi_i(p) = \int_0^p \frac{q_i(t)}{t}\, dt, \qquad \Pi_i'(p) = \frac{q_i(p)}{p} > 0 \tag{1}$$

is strictly increasing and invertible on $[0, M_i]$. On any linear segment $q_i(p) = mp + b$ over $[a, c]$, the integral increment evaluates analytically in $O(1)$ time as $m(c - a) + b \log(c/a)$, matching the PWL integration model of `pyIAST` [2]. To avoid scanning earlier knots during query execution, our shared engine precomputes cumulative prefix integrals at all knot boundaries and evaluates $\Pi_i(p)$ via binary segment lookup plus a single local segment increment.

Write $x \in (0, 1)$ for the adsorbed mole fraction of component 1. Under Ideal Adsorbed Solution Theory (IAST), binary mixture equilibrium requires the two pure-component spreading pressures to balance at the fictitious pressures $p_1 = f_1/x$ and $p_2 = f_2/(1 - x)$ [1], [6]:

$$G_f(x) = \Pi_1(f_1/x) - \Pi_2(f_2/(1 - x)) = 0, \tag{2}$$

$$D_f = [d_\ell, d_u] = [f_1/M_1,\, 1 - f_2/M_2]. \tag{3}$$

Because neither pure-component argument may exceed its finite support limit M_i, any valid equilibrium root x^* must lie within the physical domain interval $D_f = [d_\ell, d_u]$; if $d_\ell > d_u$, the query mixture is unsupported for that candidate. On a nonempty domain D_f, differentiating $G_f(x)$ yields

$$G_f'(x) = -\frac{q_1(f_1/x)}{x^2} f_1 - \frac{q_2(f_2/(1 - x))}{(1 - x)^2} f_2 < 0. \tag{4}$$

Because $G_f(x)$ is continuous and strictly decreasing on D_f, a supported equilibrium root $x^* \in D_f$ is unique and exists if and only if the domain endpoint residuals satisfy $G_f(d_\ell) \geq 0$ and $G_f(d_u) \leq 0$ (with endpoint equality permitted). Because experimental or simulated isotherm tables are only defined up to a finite maximum pressure M_i, let $C = \min\{\Pi_1(M_1), \Pi_2(M_2)\}$ denote the candidate's maximum mutual spreading pressure and precompute the support-cap pressures $r_1 = \Pi_1^{-1}(C)$ and $r_2 = \Pi_2^{-1}(C)$. Any supported equilibrium $\Pi_1(f_1/x^*) = \Pi_2(f_2/(1 - x^*)) \leq C$ requires $f_1/x^* \leq r_1$ and $f_2/(1-x^*) \leq r_2$, i.e., $x^* \in [f_1/r_1, 1-f_2/r_2]$, which is non-empty if and only if the candidate satisfies the $O(1)$ linear support guard:

$$\frac{f_1}{r_1} + \frac{f_2}{r_2} \leq 1. \tag{5}$$

This support metadata (C, r_1, r_2) is precomputed once per candidate and shared across all evaluated baselines, isolating our root-transport contribution from basic finite-domain feasibility checking.

B. Selectivity Scores, Threshold Predicates, and Ranking Contract

Given an eligible candidate's unique equilibrium root $x^* \in D_f$, its binary adsorption selectivity is defined as

$$s_y(x^*) = \frac{1-y}{y}\frac{x^*}{1-x^*}, \tag{6}$$

which increases strictly with $x^* \in (0,1)$. Consequently, testing whether a candidate's selectivity can reach a target rank threshold $\tau > 0$ is equivalent to testing whether x^* exceeds the threshold mole fraction

$$x_\tau = \frac{\tau y}{1-y+\tau y}. \tag{7}$$

If $x_\tau > d_u$, then every supported root in D_f satisfies $x^* \le d_u < x_\tau$, immediately certifying $s_y(x^*) < \tau$. Otherwise, if $x_\tau \in D_f$, a single residual probe $G_f(x_\tau)$ decides threshold survival: by strict monotonicity ($G_f' < 0$), $G_f(x_\tau) < 0$ proves $x^* < x_\tau$ and prunes the candidate, whereas $G_f(x_\tau) > 0$ proves $x^* > x_\tau$ and tightens the root bracket for a subsequent solve from $[d_\ell, d_u]$ to $[x_\tau, d_u]$. To prevent catastrophic cancellation in $1 - x_\tau$ when y or τ is large, substituting Eq. (7) into Eq. (2) yields the algebraically equivalent direct-score residual

$$H_f(t) = \Pi_1(f_1 + f_2/t) - \Pi_2(f_2 + t f_1), \tag{8}$$

which forms both pure-component pressure arguments as positive sums directly from $t > 0$ and preserves $\operatorname{sign}(H_f(t)) = \operatorname{sign}(G_f(x_t))$ throughout the supported domain.

The exact ranking contract requires returning an unordered set of k eligible candidates whose scores dominate all remaining eligible outsiders (or all eligible candidates if fewer than k exist). This contract imposes an asymmetric soundness requirement between lower and upper bounds: any candidate used to advance the kth lower threshold τ must be proved currently eligible ($f_1/r_1 + f_2/r_2 \le 1$) and associated with a distinct candidate ID, whereas an outsider's selectivity upper bound may be evaluated *conditionally* on eligibility because an ineligible candidate is excluded regardless. All compared policies execute under this identical eligibility, tie-breaking, and failure-handling contract (Section V).

IV. Transporting Root Information

Whenever a source query $f = (f_1, f_2)$ solves a candidate's equilibrium root or rejects the candidate via a fresh threshold test, it leaves behind a certified source root enclosure $[\ell, u]$. Conditional Root-Enclosure Transport maps that source interval directly into a target root enclosure $[\widehat{\ell}, \widehat{u}]$ for a new query $f' = (f_1', f_2')$ using $O(1)$ arithmetic operations, completely bypassing spreading-pressure integral evaluations whenever the transported upper bound falls below the active top-k threshold.

A. Directional Arithmetic Root Enclosure

Consider a candidate with a valid source root enclosure $\ell \le z^* \le u$ at source partial pressures $f = (f_1, f_2)$, where $z^* \in D_f$ is the unique source equilibrium root (the enclosure endpoints ℓ, u themselves need not lie inside D_f). For a target query with partial pressures $f' = (f_1', f_2')$, define the directional partial-pressure scaling ratios $c_1 = f_1'/f_1$ and $c_2 = f_2'/f_2$, along with the affine-transformed endpoints

$$\widehat{\ell} = \min\{c_1\ell,\, 1 - c_2(1-\ell)\}, \tag{9}$$

$$\widehat{u} = \max\{c_1 u,\, 1 - c_2(1-u)\}. \tag{10}$$

Theorem 1 (Conditional Root Transport)**.** *If the target query f' has a supported equilibrium root $x'^* \in D_{f'}$, then*

$$x'^* \in [\widehat{\ell}, \widehat{u}] \cap D_{f'}. \tag{11}$$

This containment holds for any positive loading curves ($q_i(p) > 0$) and any valid source enclosure $\ell \le z^* \le u$, without requiring loading monotonicity or Lipschitz derivative bounds.

Proof. Consider first the exact source root $z^* \in D_f$. Suppose toward a contradiction that $x'^* < \min\{c_1 z^*, 1 - c_2(1-z^*)\}$. Because $x'^* > 0$ and $1 - x'^* > 0$, rearranging the two inequalities $x'^* < c_1 z^*$ and $1 - x'^* > c_2(1-z^*)$ yields opposite partial-pressure ordering on the two components:

$$\frac{f_1'}{x'^*} > \frac{f_1}{z^*}, \qquad \frac{f_2'}{1-x'^*} < \frac{f_2}{1-z^*}.$$

Since both $z^* \in D_f$ and $x'^* \in D_{f'}$ are supported roots, all four pure-component pressures lie within $[0, M_i]$. Because each spreading-pressure function Π_i is strictly increasing (Eq. (1)), we obtain $\Pi_1(f_1'/x'^*) > \Pi_1(f_1/z^*)$ and $\Pi_2(f_2'/(1-x'^*)) < \Pi_2(f_2/(1-z^*))$, which implies $G_{f'}(x'^*) > G_f(z^*) = 0$, contradicting $G_{f'}(x'^*) = 0$. Symmetrically, assuming $x'^* > \max\{c_1 z^*, 1 - c_2(1-z^*)\}$ reverses both inequalities and implies $G_{f'}(x'^*) < G_f(z^*) = 0$, again a contradiction. Finally, because both affine maps $t \mapsto c_1 t$ and $t \mapsto 1 - c_2(1-t)$ are strictly increasing for $c_1, c_2 > 0$, replacing z^* by $\ell \le z^*$ in their minimum can only decrease the lower bound to $\widehat{\ell}$, and replacing z^* by $u \ge z^*$ in their maximum can only increase the upper bound to $\widehat{u}$, establishing Eqs. (9)–(11). □

Theorem 1 has two immediate operational consequences that govern our ranking engine (Fig. 2). First, mapping the intersected target enclosure $[\ell', u'] = [\widehat{\ell}, \widehat{u}] \cap D_{f'}$ through the monotone selectivity formula $s_{y'}(x)$ (Eq. (6)) yields certified target selectivity bounds $[L_i, U_i] = [s_{y'}(\ell'), s_{y'}(u')]$ in $O(1)$ arithmetic operations, independent of the number of PWL knots m_i. Whenever a candidate holds multiple valid anchors, their transported intervals intersect to tighten $[\ell', u']$ further.

Proposition 1 (Selectivity Envelope and Width Contraction)**.** *Let $m = (\ell + u)/2$ and $w = u - \ell$ denote the midpoint and width of a source enclosure $[\ell, u]$. Before intersecting with $D_{f'}$, the transported bracket width $\widehat{w} = \widehat{u} - \widehat{\ell}$ satisfies*

$$\widehat{w} \le \max(c_1, c_2) w + |c_1 m + c_2(1-m) - 1|\,. \tag{12}$$

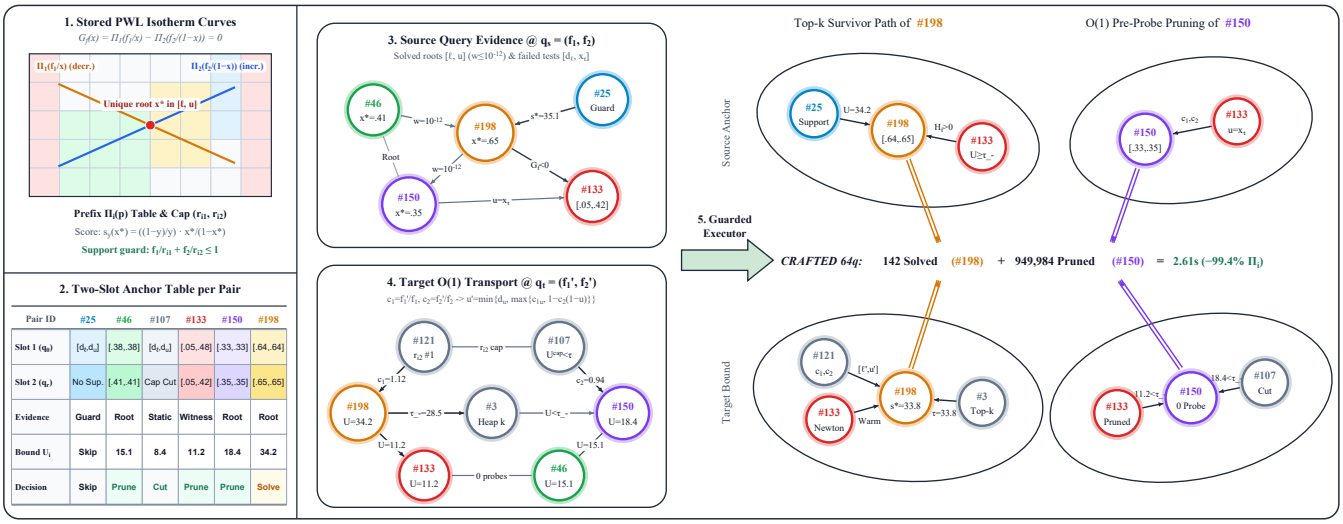


Fig. 2. **Opposite-monotone partial-pressure scaling transports both solved source roots ($w \le 10^{-12}$) and negative threshold witnesses $[d_\ell, x_\tau]$ into certified target selectivity upper bounds in $O(1)$ time, pruning 99.9% of threshold tests before any $\Pi_i(p)$ lookup. (1–2)** Each candidate pair stores precomputed piecewise-linear (PWL) spreading-pressure prefix tables $\Pi_1(p), \Pi_2(p)$, static support-cap metadata (r_{i1}, r_{i2}), and a fixed two-slot anchor table (*Slot 1: First*, *Slot 2: Recent*). **(3–4)** At a source query $q_s = (f_1, f_2)$, candidates surviving threshold testing leave tight solved root brackets $[\ell, u]$ (#198, #150, #46), whereas candidates rejected by $G_f(x_\tau) < 0$ leave one-sided negative witness brackets $[d_\ell, x_\tau]$ (#133). At a target query $q_t = (f_1', f_2')$, Theorem 1 rescales each valid anchor in $O(1)$ arithmetic via $c_1 = f_1'/f_1$ and $c_2 = f_2'/f_2$ into a target enclosure $[\ell', u']$ and selectivity upper bound U_i^{trans}. **(5 & Right)** The guarded executor scans the r_{i2}-ordered cap index, prunes outsiders with $U_i^{\text{trans}} < \tau_-$ (949,984 tests on `CRAFTED` near $k = 1$, e.g., #150, #133, with zero Π_i probes), and warm-solves only the surviving top-k contenders (142 solves, e.g., #198).

In particular, for a pure pressure step ($y' = y$, $c_1 = c_2 = c$), $\widehat{w} = cw + |c-1|$ exactly; for a pure composition step ($P' = P$), the midpoint divergence term is $|(y'-y)(m-y)/(y(1-y))|$.

Proposition 1 explains both the power and the boundary limits of root transport. When consecutive queries make moderate continuation steps ($c_1, c_2 \approx 1$) from a tightly solved source bracket ($w \approx 0$), the transported width $\widehat{w} \approx |c_1 m + c_2(1-m) - 1|$ remains an order of magnitude narrower than the static support domain $D_{f'}$, enabling aggressive upper-bound pruning. Conversely, under wide parameter jumps or near extreme feed compositions ($y \to 0$ or $y \to 1$), the midpoint divergence grows and naturally degrades the selectivity upper bound back toward the static cap bound. Specifically, for a fixed composition y and active selectivity cut $\theta > 0$ with $x_\theta = \theta y/(1 - y + \theta y)$, a source upper endpoint $u < x_\theta$ transported under pressure ratio $c = P'/P$ suffices on its own to exclude the candidate ($\max\{cu, 1 - c(1-u)\} < x_\theta$) if and only if c lies within the pressure-exclusion window

$$\frac{1 - x_\theta}{1 - u} < c < \frac{x_\theta}{u}. \tag{13}$$

B. Negative Threshold Witnesses and Two-Slot Retention

A full numerical solve on a candidate that survives threshold testing ($G_f(x_\tau) > 0$) produces an ultra-tight two-sided source root bracket $[\ell, u]$ ($w \le 10^{-12}$), which we term a *positive solved-root anchor*. Solving every candidate on every query, however, would defeat top-k pruning. Fortunately, Theorem 1 applies equally to *one-sided negative threshold witnesses* generated when a candidate fails a threshold test. Whenever a supported candidate undergoes a fresh in-domain residual probe at $x_\tau \in D_f$ and is rejected because $G_f(x_\tau) < 0$, strict monotonicity ($G_f' < 0$) certifies that its true source root z^* lies within

$$[\ell, u] = [d_\ell,\ x_\tau]. \tag{14}$$

Retaining this witness interval $[d_\ell, x_\tau]$ allows a rejected outsider to transport its upper bound $u = x_\tau$ to subsequent queries via Eq. (10) without having paid for a full root solve. By contrast, if a candidate is pruned purely by arithmetic bounds ($U_i < \tau_-$) without evaluating G_f, no new residual sign is observed and no synthetic witness is written.

To bound per-candidate memory and transport overhead in $O(1)$ space and time, each candidate maintains a fixed two-slot anchor state: a *first* slot storing the initial anchor acquired for that candidate (either from an explicit seed pass q_0 or its first encountered solve/witness), and a *recent* slot storing the anchor from the most recent query that probed the candidate. Geometrically, retaining *first* preserves a globally complete reference anchor against long-horizon drift when a serpentine or multi-pass stream returns near the initial operating region, while *recent* captures tight local continuation between consecutive queries; intersecting both transported brackets prevents interval blow-up across non-monotone trajectories. Same-query refinements intersect in place, and during target query evaluation both valid slots are transported and intersected with $D_{f'}$ before testing against the active selectivity cut.

V. Selection and Numerical Execution

Conditional root enclosures become an end-to-end query processing mechanism only when coupled with a sound top-k traversal schedule, a query-independent cap index, and a guarded floating-point execution kernel. In this section, we present the shared-kernel ranking executor that unifies all 11

evaluated policies (10 cold-pool mechanism variants plus seed-free `WarmOnDemand`) under a single correctness and lifecycle cost contract.

A. Distinct Lower Witnesses and Static Cap Indexing

Mapping a candidate's intersected target mole-fraction interval $[\ell_i, u_i] \subseteq D_{f'}$ through Eq. (6) yields a conditional selectivity interval $[L_i, U_i]$. Following the NRA minimal-probing principle [9], [10], let τ_0 denote the kth largest certified lower bound among distinct, currently eligible candidate IDs (where multiple anchors on the same candidate tighten a single interval $[L_i, U_i]$ and count at most once). Whenever k such eligible witnesses exist, at least k candidates are guaranteed to achieve selectivity at least τ_0, so any candidate whose conditional upper bound satisfies $U_i < \tau_0$ is provably excluded from the top-k answer without probing its target spreading-pressure integrals.

To order candidate traversal without materializing $O(N)$ transported bounds upfront, we exploit the shared support-cap metadata (r_{i1}, r_{i2}) from Eq. (5). Because any supported equilibrium satisfies $\Pi_{i1}(p_1) = \Pi_{i2}(p_2) \le C_i$, the pure-component partial pressures satisfy $f_1/x^* \le r_{i1}$ and $f_2/(1-x^*) \le r_{i2}$, which restricts x^* to the query-local cap bracket

$$x^* \ge \frac{f_1}{r_{i1}}, \qquad x^* \le 1 - \frac{f_2}{r_{i2}}. \tag{15}$$

Mapping Eq. (15) through the selectivity definition (Eq. (6)) produces closed-form query-local cap score bounds:

$$U_i^{\mathrm{cap}} = \frac{r_{i2} - f_2}{f_1}, \qquad L_i^{\mathrm{cap}} = \frac{f_2}{r_{i1} - f_1}. \tag{16}$$

Crucially, for any query (f_1, f_2), U_i^{cap} is a strictly increasing affine function of r_{i2}, and L_i^{cap} (among eligible candidates with $r_{i1} > f_1$) is a strictly decreasing function of r_{i1}. Consequently, sorting the candidate pool once in descending order of r_{i2} (and ascending order of r_{i1} for lower-bound seeding) provides a *query-independent static cap index*. During query execution, as soon as the scanning cursor reaches a candidate whose static cap bound satisfies $U_i^{\mathrm{cap}} < \tau$, the entire remaining suffix of the r_{i2}-ordered index is eliminated in $O(1)$ time without even checking individual candidate eligibility.

Indexed query procedure. Input: query $z = (P, y, k)$ with $k > 0$; guarded threshold cut τ_- defined by Eq. (17).

1) **Lower-Bound Seeding:** Scan the r_{i1}-ordered lower-seed cursor (and previous query's top-k incumbents when enabled) to find up to k eligible distinct candidates, tightening lower bounds via transported anchors when enabled. Initialize τ from their kth largest certified lower bound (or 0 if fewer than k exist).
2) **Cap-Ordered Traversal & Suffix Cut:** Visit remaining candidates in descending r_{i2} static cap order, skipping already-visited IDs. As soon as $U_i^{\mathrm{cap}} < \tau_-$, terminate the scan in $O(1)$ time (eliminating the entire remaining suffix).
3) **Support Guard & Bound Transport:** Check the linear support guard $f_1/r_{i1} + f_2/r_{i2} \le 1$ (Eq. (5)); skip unsupported candidates, or escalate to the 112-bit `binary128` fallback check if interval rounding straddles 1. Intersect $D_{f'}$ with transported anchor brackets (Eqs. (9)–(10)) to form $[L_i, U_i]$.
4) **Pre-Probe Upper-Bound Pruning:** If $U_i < \tau_-$, prune the candidate immediately with zero $\Pi_i(p)$ integral probes (without writing a new source witness).
5) **Guarded Residual Threshold Test:** For surviving candidates ($U_i \ge \tau_-$), evaluate the outward-rounded interval residual $H_f(\tau_-)$ (Eq. (8)). If strictly negative, reject the candidate and (when witness retention is enabled) store $[d_\ell, x_{\tau_-}]$; if strictly positive, tighten the solve bracket to $[x_{\tau_-}, u_i]$; if indeterminate around zero, clear uncertified threshold hints and proceed to Step 6.
6) **Warm-Started Root Solve & Anchor Update:** Solve for $x^* \in [\ell_i, u_i]$ using safeguarded Newton–bisection (escalating to the exact `binary128` fallback solver if fast floating-point brackets do not reach the 10^{-12} target width). Retain the solved bracket in the anchor cache when enabled and update the top-k heap and threshold τ.
7) **Tolerance-Shell Canonicalization:** Form the boundary tolerance shell $[t - m(t), t + m(t)]$, trigger canonical fallback refinement on any near-tied strict winners, sort deterministically, and return the top-k set.

As summarized in the indexed execution procedure above, each query proceeds in three coordinated stages that generalize classic NRA and minimal-probing top-k execution [9], [10] to implicit equilibrium scores: (1) *Threshold Initialization*, which seeds τ from the r_{i1} lower-bound order and evaluates the previous query's top-k incumbents; (2) *Cap-Ordered Traversal*, which scans candidates in descending r_{i2} order, terminates early on suffix cap cuts ($U_i^{\mathrm{cap}} < \tau_-$), and applies $O(1)$ root transport to visited candidates to prune those with $U_i^{\mathrm{trans}} < \tau_-$ before any residual probe; and (3) *Guarded Probing and Retention*, which tests surviving candidates via $H_f(\tau_-)$, solves positive survivors via warm-started Newton–bisection, and updates the two-slot anchor cache. When evaluating pruning attribution in Section VII, an upper-bound exclusion ($U_i < \tau_-$) is classified as *cap-sufficient* if the static query-local cap bound alone would have pruned the candidate ($U_i^{\mathrm{cap}} < \tau_-$), and *history-needed* if $U_i^{\mathrm{cap}} \ge \tau_-$ but the transported historical enclosure tightens the upper bound below the cut ($U_i^{\mathrm{trans}} < \tau_-$). All ten cold-pool mechanism policies (`Incumbent`, `Cap--Heap`, `Cap--Index`, `LeanCap`, `Seed--Cap`, `Seed--Warm`, `Roots--Only`, `First--Only`, `Transport`, and `Heap--Transport`) plus seed-free `WarmOnDemand` execute within this identical C++20 framework.

B. Guarded Floating-Point Predicates and Tolerance Shell

To ensure that finite-precision rounding never fabricates a false exclusion or corrupts a retained anchor, all policies share an explicit numerical guard contract. At the target relative/absolute score tolerance 10^{-12}, let t denote the kth score in the raw top-k min-heap and define the boundary tolerance

shell $m(t) = \max(10^{-10}, 10^{-8}|t|)$. Candidates with scores strictly above $t + m(t)$ are strict winners, while candidates within $[t - m(t), t + m(t)]$ belong to the boundary shell and are ordered deterministically by score and ascending candidate ID (with cold canonical re-solves triggered if adjacent strict winners differ by less than $32 \times 10^{-12} \max(1, |s_i|, |s_j|)$). To guarantee that arithmetic and predicate pruning preserve every member of the true boundary shell, each lower threshold τ is widened downward by $(1 - 1024\varepsilon_{\mathrm{mach}})$ and tested against the guarded cut

$$\tau_- = \tau - 1.001 \max(10^{-10}, 10^{-8}|\tau|). \tag{17}$$

Because $c(u) = u - 1.001m(u)$ is strictly increasing for $u > 0$, any valid lower threshold $\tau \le t^*$ satisfies $\tau_- = c(\tau) \le c(t^*) < t^* - m(t^*)$, ensuring that no candidate inside the true tolerance shell can be pruned by $U_i < \tau_-$.

Every interval operation—including prefix integrals, pressure ratios c_1, c_2, transport maps (Eqs. (9)–(10)), and direct-score residuals $H_f(\tau_-)$ (Eq. (8))—uses outward-rounded 'long double' interval arithmetic (widening basic operations by adjacent machine numbers and `log1p` evaluations by 4 ulp). A threshold probe $H_f(\tau_-)$ rejects a candidate (and records a negative witness $[d_\ell, x_{\tau_-}]$) only when the entire residual interval is strictly negative; if the interval contains zero or encounters uncertain domain support, the predicate returns *unknown*, discards uncertified witness hints, and falls through to the numerical solver. The fast solver runs safeguarded Newton–Raphson (with bisection every 8th step) initialized from the transported bracket (if endpoint residual signs oppose) or Henry's-law slope, escalating to a 112-bit 'binary128' fallback solver (`Curve::precise_pi` and `solve_precise`) whenever fast brackets fail to meet the 10^{-12} score-width target.

C. Complexity and Paid Lifecycle Accounting

For a pool of N candidates with at most two anchors per candidate, building the static cap index costs $O(N \log N)$ once at engine construction, and all retained index/anchor structures occupy $O(N)$ memory (2× two-slot anchors add 320 B per candidate). On each query, scanning s lower-seed candidates, visiting v cap-ordered candidates, and maintaining the top-k heap across b solved survivors requires $O(s + v + k + b \log(k + 1))$ index overhead plus the cost of residual and solver probes. Because transported upper bounds avoid both Π_1 and Π_2 prefix evaluations on excluded candidates, their net runtime benefit depends on whether cumulative probe savings exceed initial acquisition cost. For a stream of m queries with constructor, cap-index, and seed-acquisition cost A, per-query execution costs C_j, and teardown cost D, the total *paid lifecycle time* is

$$T(m) = A + \sum_{j=1}^{m} C_j + D. \tag{18}$$

If seed acquisition adds upfront overhead $\Delta A > 0$ over an unseeded baseline while saving an average of $\delta > 0$ per query, break-even occurs once $m\delta > \Delta A$. Our evaluation measures $T(m)$ directly across prefixes $m \in \{1, 8, 16, 32, 64\}$ so that every crossover horizon is observed empirically under full lifecycle accounting.

VI. Experimental Setup

We evaluate conditional root-enclosure transport through a three-part experimental program designed to answer four systems questions: (1) whether transporting historical root enclosures reduces end-to-end paid lifecycle cost once constructor, index-build, and seed-acquisition overheads are charged; (2) which specific algorithmic component—static cap bounds, solver warm starts, positive solved-root anchors, or negative threshold witnesses—drives candidate pruning; (3) how seeded root transport compares against a practical seed-free `WarmOnDemand` baseline across realistic CO_2 capture pressures; and (4) where the exact operating boundaries lie with respect to query horizon m, pool size N, ranking depth k, and query-to-query jump width.

A. Benchmark Relations and Workload Regimes

Our evaluation spans curated single-component adsorption isotherms from NIST ISODB [3], [5] and simulated metal-organic framework (MOF) isotherms from CRAFTED [4]. Every isotherm is represented as a piecewise-linear (PWL) loading curve in $(\mathrm{bar}, \mathrm{mmol/g})$ with precomputed exact segment slopes and prefix spreading-pressure integrals $\Pi_i(p)$. Candidate pairs are formed within the same material under a fixed temperature window, yielding three primary benchmark relations:

- **`CRAFTED` CO_2/N_2 ($N = 16{,}010$ pairs):** The primary large-pool screening relation over 290–310 K, used across all cold-pool mechanism, warm-pool continuation, scaling, and arrival-regime experiments.
- **`NIST--N2` CO_2/N_2 ($N = 353$ pairs):** The primary experimental flue-gas relation over 293–303 K, used in the cold-pool mechanism matrix, the literature operating-range pressure grid, and arrival diagnostics.
- **`NIST--CH4` CO_2/CH_4 ($N = 373$ pairs):** A complementary experimental biogas/natural-gas upgrading relation over 293–303 K, used in the continuation-regime `WarmOnDemand` cohort.

To separate local continuation from wider parameter sweeps, our primary mechanism benchmark evaluates two 64-query stream regimes at ranking depths $k \in \{1, 50\}$ following a common initial seed acquisition query ($P_0 = 0.3$ bar, $y_0 = 0.15$): a **near continuation regime** that sweeps $P \in [0.3, 1.4]$ bar and $y \in [0.12, 0.28]$ in serpentine grid order (median step $\Delta P = 0.0$ bar, $\Delta y = 0.0229$), and a **far continuation regime** that sweeps $P \in [0.1, 4.0]$ bar and $y \in [0.02, 0.70]$ with larger parameter jumps (median step $\Delta P = 0.0$ bar, $\Delta y = 0.0971$). Prefix crossover behavior is measured at horizons $m \in \{1, 8, 16, 32, 64\}$ by running standalone paid lifecycles on the first m stream queries.

B. Compared Baselines and Ablation Design

To ensure a fair systems comparison free of implementation bias, all compared methods execute inside the same single-threaded C++20 engine (compiled with `g++ -O3 -std=c++20`), sharing identical PWL isotherm tables, prefix spreading-pressure arrays, linear support guards, top-k heap structures, and safeguarded Newton–bisection root solvers (10^{-12} target score tolerance with 112-bit `binary128` fallback). Our evaluation compares three methodological families:

- **Query-Local Cap Indexing and Continuation Baselines:** `Incumbent` evaluates the previous query's top-k winners first to raise τ early before scanning remaining candidates via direct-score residual tests $H_f(\tau_-)$. `Cap--Heap` constructs a per-query priority queue over U_i^{cap} to pop candidates in descending cap order until $U_i^{\mathrm{cap}} < \tau_-$. `LeanCap` replaces per-query heap construction with our query-independent r_{i2}-sorted static index and $O(1)$ suffix cutoff (`Cap--Index` additionally allocates an empty two-slot anchor buffer to verify that memory allocation alone adds negligible overhead).
- **Solver-Only Warm-Starting Controls:** `Seed--Warm` pays the initial seed query q_0 at (P_0, y_0), retains solved roots, and uses transported brackets exclusively to warm-start the numerical solver *after* a candidate survives $H_f(\tau_-)$, without upper-bound pruning (`Seed--Cap` pays q_0 without solver warming to isolate pure seed-phase cost). `WarmOnDemand` eliminates the upfront seed query q_0: it starts with an empty cache, uses static cap bounds for pruning, and caches positive roots on demand as candidates are solved.
- **Certified Root-Enclosure Transport (Ours):** `Transport` combines the static r_{i2} cap index with two-slot (*first* + *recent*) retention of both positive solved roots and negative threshold witnesses for $O(1)$ pre-probe upper-bound pruning and warm solver initialization. To isolate each design choice, `Roots--Only` retains only positive solved roots (omitting negative witness writes), `First--Only` retains only a single immutable *first* slot per pair, and `Heap--Transport` pairs two-slot transport with a dynamic per-query cap heap in place of the static r_{i2} index.

C. Operating-Range Generalization and Scaling Diagnostics

Beyond the near and far continuation streams, we evaluate how root-enclosure transport generalizes across physical operating regimes, repository scales, and stochastic query arrivals. First, in the **Multi-Order Operating-Pressure Sweep** (Table IV, Part B), we test `NIST--N2` and `CRAFTED` across two orders of magnitude in pressure ($P \in \{0.01, 0.05, 0.1, 1.0\}$ bar) and CO_2 mole fractions $y \in \{0.05, 0.10, 0.15, 0.50\}$, covering both atmospheric flue-gas capture (0.1–1 bar) and dilute modular scrubbing (0.01 bar), comparing policies seeded at either $P_0 = 0.01$ bar or $P_0 = 1.0$ bar ($y_0 = 0.15$) against seed-free `WarmOnDemand`. Second, in the **Scaling and Structural Boundary Suite** (Fig. 4a, Tables II–V), we sweep candidate pool sizes $N \in \{256, 1{,}024, 4{,}096, 16{,}010\}$ and ranking depths $k \in \{1, 10, 50, 200\}$, measure eligible support rates and normalized rank-k selectivity gaps $(s_k - s_{k+1})/\max(1, |s_k|)$, and evaluate 16-query stochastic streams under independent normal arrival perturbations (`S1`–`S3`) at both *small* and *wide* jump variances.

D. Paid Lifecycle Accounting and Verification Protocol

Every timing measurement in this paper enforces a strict **paid lifecycle contract** (Eq. (18)): the wall-clock timer starts prior to policy construction, charges static cap-index sorting, charges the full execution of any declared seed acquisition query q_0 at (P_0, y_0), executes the m stream queries, and stops only after object teardown. Fine-grained attribution counters (such as history-certified vs. static-cap exclusions and query-phase $\Pi_i(p)$ probe counts) are collected in separate diagnostic runs so instrumentation never perturbs timed runs. Every experiment is repeated across three randomized paired blocks, reporting the median and full $[\min, \max]$ range. Across all timed and diagnostic runs, every policy achieved **zero top-k ID mismatches** and zero score deviations exceeding 10^{-9} relative to an exhaustive root-solving oracle.

VII. Experimental Results

We organize our empirical findings around the four systems questions posed in Section VI: (1) paid lifecycle performance across the 10-policy cold-pool mechanism matrix and prefix crossover horizons (Section VII-A); (2) mechanism attribution isolating bound pruning from solver warm-starting (Section VII-B); (3) comparisons against seed-free `WarmOnDemand` across near/far streams and the 0.01–1 bar literature operating-range grid (Section VII-C); and (4) scaling, descriptor, and arrival-width operating boundaries (Section VII-D).

A. Paid Cold-Pool Lifecycle and Acquisition Crossover

Root-enclosure transport cuts 64-query paid cold-pool time by 37%–54% vs. `LeanCap` and 26%–50% vs. `Incumbent` on large repositories. Table I reports the full 10-policy cold-pool lifecycle times at 64 queries, while Table II decomposes the paid lifecycle into native construction, seed acquisition, query execution, and teardown alongside engine-owned memory. On `CRAFTED` ($N = 16{,}010$), `Transport` completes the 64-query lifecycle in 2.61 s (near $k = 1$) and 2.55 s (near $k = 50$), and in 3.15 s (far $k = 1$) and 3.61 s (far $k = 50$), reducing paid wall-clock time to 0.46×–0.63× of query-local cap indexing (`LeanCap`, 5.14–5.72 s) and 0.50×–0.74× of incumbent continuation (`Incumbent`, 4.90–5.09 s). Even though `Transport` pays a one-time seed phase (141.11 ms on `CRAFTED` and 2.58 ms on `NIST--N2` in Table II, with 2,501.6 KiB and 55.2 KiB of anchor storage, respectively) to solve all eligible pairs during q_0, it cuts the 64-query execution time by 2.3×–4.5× over `LeanCap`, easily amortizing the initial seed investment. On the smaller experimental `NIST--N2` pool ($N = 353$), `Transport` outperforms unseeded baselines across all three

TABLE I

CERTIFIED ROOT-ENCLOSURE TRANSPORT CUTS 64-QUERY PAID COLD-POOL RUNTIME BY 37%–54% ON CRAFTED AND WINS 7 OF 8 BENCHMARK CELLS, WHEREAS SOLVER-ONLY WARM-STARTING (SEED--WARM) IS SLOWER THAN UNSEEDED BASELINES. ENTRIES REPORT MEDIANS (SECONDS) ACROSS THREE RANDOMIZED BLOCKS UNDER THE PAID LIFECYCLE CONTRACT (CHARGING CURVE SETUP, STATIC-INDEX CONSTRUCTION, DECLARED SEED ACQUISITION q_0, 64 STREAM QUERIES, AND TEARDOWN). **COLOR & TYPOGRAPHY LEGEND:** RELATIVE TO INCUMBENT, **TEAL** ↓ DENOTES FASTER IN ALL THREE BLOCKS, AMBER ∼ DENOTES MIXED OUTCOMES, AND COPPER ↑ DENOTES SLOWER IN ALL THREE BLOCKS; THE LOWEST MEDIAN PAID TIME PER ROW IS UNDERLINED.

			Unseeded Cap & Continuation Baselines				Seeded Controls		Certified Root-Enclosure Transport (Ours)			
Relation	Regime	k	Incumbent	Cap–Heap	Cap–Index	LeanCap	Seed–Cap	Seed–Warm	Roots–Only	First–Only	Transport	Heap–Trans.
NIST–N_2	near	1	0.0640	0.0786↑	0.0614∼	0.0560∼	**0.0547**↓	**0.0576**↓	**0.0399**↓	**0.0399**↓	**0.0417**↓	**0.0567**↓
NIST–N_2	near	50	0.0946	0.1406↑	0.1221↑	0.1189↑	0.1205↑	0.1268↑	**0.0903**↓	0.0886∼	**0.0890**↓	**0.0881**↓
NIST–N_2	far	1	0.0660	0.0704∼	**0.0611**↓	**0.0609**↓	**0.0636**↓	0.0710∼	**0.0472**↓	**0.0505**↓	**0.0511**↓	**0.0600**↓
NIST–N_2	far	50	0.0908	0.1414↑	0.1109↑	0.1226↑	0.1328↑	0.1277↑	0.1194↑	0.1196↑	0.1263↑	0.1269↑
CRAFTED	near	1	4.9496	5.6670↑	5.2479↑	5.2196∼	5.3186↑	5.6033↑	**2.4741**↓	**2.6367**↓	**2.6127**↓	**2.5468**↓
CRAFTED	near	50	5.0891	6.1336↑	5.3282∼	5.4823↑	5.4862∼	5.9882↑	**2.5255**↓	**2.5605**↓	**2.5477**↓	**2.6762**↓
CRAFTED	far	1	4.8992	5.9022↑	5.1612∼	5.1353∼	5.6349↑	5.8535↑	**3.0149**↓	**3.0625**↓	**3.1525**↓	**3.2343**↓
CRAFTED	far	50	4.8981	6.4371↑	5.7606↑	5.7203↑	5.7187↑	6.7551↑	**3.4831**↓	**3.6469**↓	**3.6085**↓	**3.6615**↓

TABLE II

ONE-TIME SEED ACQUISITION (2.58–141.11 MS) AND TWO-SLOT ANCHOR STORAGE (55.2–2,501.6 KIB) CUT 64-QUERY PAID RUNTIME BY 26%–50% (NEAR $k = 1$). MEDIANS IN MS; **SOFT TEAL** HIGHLIGHTS TRANSPORT.

Relation	Policy	Paid (ms)	Build (ms)	Seed q_0 (ms)	Free (ms)	Owned (MiB)	Anchors (KiB)	Valid slots
NIST–N_2	LeanCap	55.99	22.48	0.01	0.45	2.516	0.0	0
NIST–N_2	Cap–Index	61.36	21.50	0.01	0.46	2.586	55.2	0
NIST–N_2	**Transport**	**41.67**	21.55	2.58	0.40	2.586	55.2	312
CRAFTED	LeanCap	5219.56	656.08	0.03	13.75	65.268	0.0	0
CRAFTED	Cap–Index	5247.91	624.33	0.03	14.27	68.444	2501.6	0
CRAFTED	**Transport**	**2612.74**	636.33	141.11	14.35	68.444	2501.6	15498

TABLE III

HIGH SUPPORT ELIGIBILITY (85.4%–96.1%) LEAVES STATIC CAPS WIDE, WHILE TRANSPORTED BRACKETS RESOLVE TIGHT RANK-k BOUNDARY GAPS $(s_k - s_{k+1})/\max(1, |s_k|)$. **TEAL** = PAID WIN; COPPER = HIGH-k LOSS.

Relation	Regime	k	Support	Gap (q_1)	Gap (q_{64})
NIST–N_2	near	1	87.5%	0.38	0.37
NIST–N_2	near	50	87.5%	0.0024	0.0025
NIST–N_2	far	1	85.4%	0.32	0.39
NIST–N_2	far	50	85.4%	0.0084	0.012
CRAFTED	near	1	93.2%	5.4×10^{-4}	4.6×10^{-6}
CRAFTED	near	50	93.2%	1.6×10^{-3}	7.9×10^{-4}
CRAFTED	far	1	96.1%	2.5×10^{-3}	4.0×10^{-3}
CRAFTED	far	50	96.1%	5.5×10^{-4}	3.7×10^{-3}

selective configurations (near $k = 1$: 0.0417 s vs. 0.0640 s for `Incumbent` and 0.0560 s for `LeanCap`; near $k = 50$: 0.0890 s vs. 0.0946 s and 0.1189 s; far $k = 1$: 0.0511 s vs. 0.0660 s and 0.0609 s), and trails only in the high-return far $k = 50$ regime (0.1263 s vs. 0.0908 s for `Incumbent`) where $k/N = 14.2\%$ of the repository must be returned on every query.

Seed acquisition breaks even by $m^* = 8$ queries on near streams and $m^* = 16$ queries on far streams over CRAFTED. Because any seeded policy pays a one-time cold-start solve at q_0, single-query lifecycles ($m = 1$) do not amortize seed acquisition against unseeded baselines (Fig. 1b and Fig. 3a). However, as the query horizon m grows, cumulative savings from skipped integral probes rapidly overtake the seed cost. As shown by the break-even horizon annotations m^* in Fig. 3a and the radar speedup profile in Fig. 1c, on `CRAFTED`, `Transport` overtakes `Incumbent`, `LeanCap`, and `Seed--Warm` across all blocks by $m^* = 8$ queries in both near configurations ($k = 1, 50$) and by $m^* = 16$ queries against `Incumbent` in both far configurations ($m^* = 16$–32 vs. `LeanCap`; $m^* = 1$ vs. `Seed--Warm`). On `NIST--N2`, break-even against `Incumbent` occurs at $m^* = 16$ (near $k = 1$, far $k = 1$) and $m^* = 32$ (near $k = 50$). Once break-even is reached, the advantage is monotone and persistent across all remaining prefix horizons ($m = 16, 32, 64$).

B. Mechanism Attribution: Bound Transport vs. Solver Warming

Warming the numerical solver without transporting upper bounds fails to reduce candidate probes and loses to unseeded baselines. A central hypothesis of our work is that historical roots yield their primary systems value by pruning candidates *before* solver entry rather than accelerating Newton iterations *inside* the solver. Comparing `Transport (T)` against `Seed--Warm (S--W)` in Table I and Fig. 3a confirms this hypothesis decisively. Both policies pay the exact same seed acquisition cost at q_0, yet `Seed--Warm` requires 5.60–6.76 s on `CRAFTED`—slower than stateless `LeanCap` (5.14–5.72 s)—because it still executes 1.91M–2.17M query-phase $\Pi_i(p)$ integral probes across the 64 queries (nearly identical to `LeanCap`'s 1.91M–2.22M probes). By contrast, `Transport` converts the same seed brackets into target-query upper bounds, slashing query-phase $\Pi_i(p)$ probes by 78.3%–99.4% down to 0.011M–0.481M and cutting paid lifecycle time to 0.43×–0.54× of `Seed--Warm`.

Transported historical brackets shrink target root intervals by up to 32.8× and drive 99.9% of all upper-bound exclusions on CRAFTED. Fig. 3b–c explains why static cap

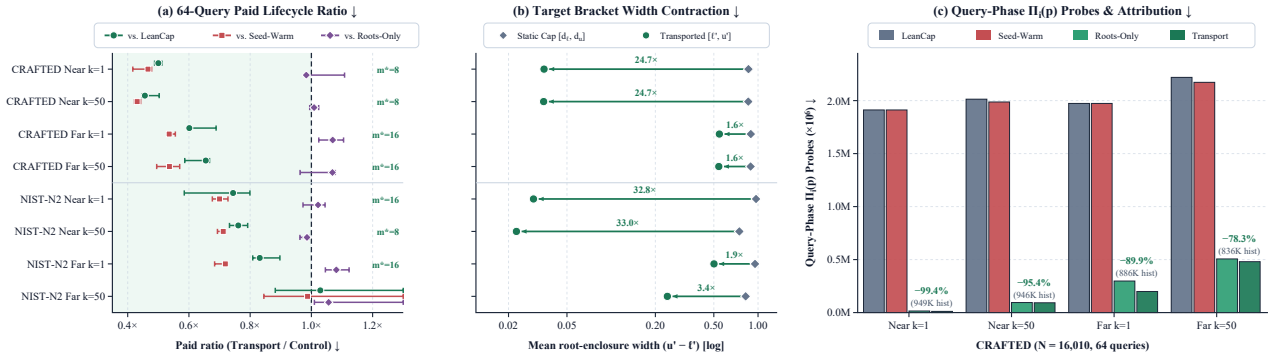


Fig. 3. **Transported historical brackets shrink target root intervals by up to** $32.8\times$ **and drive** 99.9% **of all upper-bound exclusions on** `CRAFTED`**, whereas solver warm-starting without bound transport (**`Seed--Warm`**) fails to prune candidates and loses to stateless indexing. (a)** 64-query paid lifecycle ratio forest plot (`Transport` / Control, median and 3-block $[\min, \max]$ whiskers) across `CRAFTED` and `NIST--N2` against `LeanCap`, `Seed--Warm`, and `Roots--Only`, annotated with the exact prefix break-even horizon m^* vs. `Incumbent` / `LeanCap`; values left of $1.0\times$ denote paid speedups. **(b)** Mean certified target root-enclosure width contraction from static query-local cap intervals (`LeanCap`) to transported historical brackets (`Transport`), achieving $24.6\times$ contraction on `CRAFTED` Near ($0.861 \to 0.035$) and $32.8\times$ on `NIST--N2` Near ($0.970 \to 0.030$). **(c)** Query-phase upper-bound exclusion attribution separating history-certified rejections from static cap-sufficient rejections on `CRAFTED`, alongside query-phase $\Pi_i(p)$ integral probe reductions (-78.3% to -99.4%).

bounds alone cannot prune implicit equilibrium scores and why root transport succeeds. Under `LeanCap`, query-local support caps leave mean root enclosures of width 0.858–0.893 on `CRAFTED` and 0.746–0.970 on `NIST--N2`. Transporting source brackets via Theorem 1 contracts those mean intervals to 0.035 on `CRAFTED` near ($24.6\times$ narrower) and 0.541–0.545 on `CRAFTED` far ($1.64\times$–$1.65\times$ narrower), and to 0.023–0.030 on `NIST--N2` near ($32.8\times$–$33.0\times$ narrower; $116\times$ median contraction from 0.931 to 0.008) and 0.241–0.501 on `NIST--N2` far ($1.90\times$–$3.41\times$ narrower). Diagnostic attribution in Fig. 3c reveals that in `CRAFTED` near $k = 1$ and $k = 50$, static cap bounds suffice for only 704 candidate exclusions, whereas 949,984 and 946,251 exclusions (99.93%) strictly require transported historical brackets ('history-needed'). Even in the wider far regime, transported history accounts for 886,979 ($k = 1$) and 836,360 ($k = 50$) exclusions vs. only 704 and 653 cap-sufficient exclusions.

Static cap indexing outperforms dynamic heap ordering, while positive-root-only retention (`Roots--Only`**) offers a fast, lightweight default.** Across the ablation ladder in Table I, replacing a per-query cap priority queue (`Cap--Heap`, 5.67–6.44 s on `CRAFTED`) with our static r_{i2}-sorted index (`Cap--Index`, 5.16–5.76 s) saves 0.42–0.81 s of heap overhead, while pairing two-slot transport with the static index (`Transport`, 2.55–3.61 s) edges out dynamic heap transport (`Heap--Transport`, 2.55–3.66 s) in 3 of 4 `CRAFTED` cells and all selective $k = 1$ `NIST--N2` cells. Freezing retention after the first slot (`First--Only`, 2.56–3.65 s on `CRAFTED`) trails two-slot `Transport` when repeated probes refine brackets across the stream (199K vs. 286K query Π_i probes on `CRAFTED` far $k = 1$). Meanwhile, `Roots--Only (R)` (retaining only positive solved roots and skipping negative witness writes) is 0%–8% faster than full `Transport` in 7 of 8 cold-pool cells (2.47–3.48 s on `CRAFTED`), demonstrating that positive solved roots from q_0 carry the vast majority of pruning power, whereas negative threshold witnesses further cut query-phase Π_i probes (e.g., from 297.7K under `Roots--Only` to 199.1K under `Transport` on `CRAFTED` far $k = 1$, and 0.0890 s vs. 0.0903 s on `NIST--N2` near $k = 50$).

C. Comparison Against Seed-Free WarmOnDemand and Pressure Grid

Against a practical seed-free `WarmOnDemand` **baseline, root-enclosure transport consistently outperforms across all eight continuation-regime streams at 64 queries.** A natural systems question is whether an unseeded engine that simply caches roots on demand as candidates are solved (`WarmOnDemand`) can match seeded root transport without paying for q_0. Part A of Table IV answers this directly across `NIST--CH4` and `CRAFTED`. Because `WarmOnDemand` uses cached roots only to warm-start the solver and relies on static cap bounds for pruning, it must still probe nearly every candidate on every query. Consequently, `Transport` achieves consistent paid speedups across all 8 Part A configurations at 64 queries, reducing lifecycle time to $0.485\times$–$0.623\times$ of `WarmOnDemand` on `CRAFTED` (breaking even by $m^* = 8$ on near and $m^* = 16$ on far) and $0.617\times$–$0.961\times$ on `NIST--CH4`. In contrast, `Seed--Warm` relative to `WarmOnDemand` exceeds $1.07\times$–$1.39\times$ across every row, proving that paying for a seed query without bound transport is strictly harmful relative to on-demand caching.

Across the 0.01**–**1 **bar literature pressure grid,** 1 **bar seed acquisition wins decisively on** `CRAFTED`**, whereas** 0.01 **bar cold starts and high-**k **small pools favor** `WarmOnDemand`**.** Part B of Table IV and Fig. 4b–c expose an important physical operating boundary when sweeping across two orders of magnitude in pressure ($P \in \{0.01, 0.05, 0.1, 1.0\}$ bar). When seeded at flue-gas atmospheric pressure ($P_0 = 1$ bar), `Transport` and `Roots--Only` achieve strong paid speedups on `CRAFTED` ($0.609\times$–$0.724\times$ for `Transport`; $0.560\times$–$0.609\times$ for `Roots--Only`, breaking even by $m^* = 8$–16 queries) and

TABLE IV

SEEDED ROOT-ENCLOSURE TRANSPORT OVERTAKES SEED-FREE WARMONDEMAND ACROSS ALL EIGHT CONTINUATION STREAMS (BY $m^* = 8$–16 ON CRAFTED) AND 1 BAR PRESSURE SWEEPS, WHILE 0.01 BAR COLD STARTS AND HIGH-k SMALL POOLS FAVOR WARMONDEMAND. ENTRIES REPORT 64-QUERY MEDIAN [MINIMUM, MAXIMUM] PAID LIFECYCLE RATIOS ACROSS THREE PAIRED BLOCKS, ALONGSIDE THE BREAK-EVEN QUERY HORIZON $m^* \in \{1, 8, 16, 32, 64\}$ (AND 5-HORIZON STATUS SEQUENCE) WHERE SEEDED TRANSPORT OVERTAKES WARMONDEMAND. **COLOR LEGEND:** TEAL ↓ / W = FASTER IN ALL THREE BLOCKS; AMBER ∼ / M = MIXED; COPPER ↑ / L = SLOWER IN ALL THREE BLOCKS.

Relation	Regime / Seed P_0	k	64-Query Paid Lifecycle Ratio vs. WarmOnDemand			Break-Even m^* ($m = 1, 8, 16, 32, 64$)	
			Transport (Ours)	Roots--Only (Ours)	Seed--Warm	Transport (Ours)	Roots--Only (Ours)
Part A: Near and far continuation streams (seeded policies initialize anchors at $P_0 = 0.3$ bar, $y_0 = 0.15$)							
NIST–CH_4	near	1	**0.617 [0.571,0.782]**$^{\downarrow}$	**0.631 [0.591,0.857]**$^{\downarrow}$	1.388 [0.856,2.034]$^{\sim}$	m^*=16 (MMWWW)	m^*=32 (MWMWW)
NIST–CH_4	near	50	**0.767 [0.604,0.872]**$^{\downarrow}$	0.824 [0.602,1.080]$^{\sim}$	1.178 [0.784,1.497]$^{\sim}$	m^*=64 (LMMMW)	– (MMMMM)
NIST–CH_4	far	1	**0.833 [0.773,0.834]**$^{\downarrow}$	**0.801 [0.789,0.968]**$^{\downarrow}$	1.235 [1.160,1.252]$^{\uparrow}$	m^*=32 (LLMWW)	m^*=64 (LLLMW)
NIST–CH_4	far	50	**0.961 [0.955,0.961]**$^{\downarrow}$	1.083 [0.971,1.097]$^{\sim}$	1.074 [0.992,1.123]$^{\sim}$	m^*=32 (MMMWW)	– (LLLMM)
CRAFTED	near	1	**0.485 [0.464,0.534]**$^{\downarrow}$	**0.473 [0.455,0.527]**$^{\downarrow}$	1.130 [1.121,1.132]$^{\uparrow}$	m^*=8 (LWWWW)	m^*=8 (LWWWW)
CRAFTED	near	50	**0.492 [0.487,0.501]**$^{\downarrow}$	**0.486 [0.462,0.501]**$^{\downarrow}$	1.136 [1.133,1.157]$^{\uparrow}$	m^*=8 (LWWWW)	m^*=8 (LWWWW)
CRAFTED	far	1	**0.584 [0.567,0.620]**$^{\downarrow}$	**0.589 [0.486,0.595]**$^{\downarrow}$	1.122 [1.038,1.139]$^{\uparrow}$	m^*=16 (MMWWW)	m^*=16 (MMWWW)
CRAFTED	far	50	**0.623 [0.603,0.639]**$^{\downarrow}$	**0.585 [0.554,0.593]**$^{\downarrow}$	1.144 [1.063,1.156]$^{\uparrow}$	m^*=16 (LMWWW)	m^*=16 (LMWWW)
Part B: Multi-order operating-pressure sweep ($P \in [0.01, 1.0]$ bar; seeded policies initialize at $P_0 \in \{0.01, 1.0\}$ bar, $y_0 = 0.15$)							
NIST–N_2	$P_0 = 0.01$ bar	1	**0.950 [0.795,0.956]**$^{\downarrow}$	**0.740 [0.720,0.982]**$^{\downarrow}$	1.073 [0.787,1.124]$^{\sim}$	m^*=1 (WWWWW)	m^*=1 (WWWWW)
NIST–N_2	$P_0 = 0.01$ bar	50	1.099 [1.079,1.248]$^{\uparrow}$	1.164 [1.046,1.212]$^{\uparrow}$	1.054 [1.049,1.126]$^{\uparrow}$	– (MMLLL)	– (LLLLL)
NIST–N_2	$P_0 = 1.0$ bar	1	**0.942 [0.875,0.972]**$^{\downarrow}$	**0.867 [0.791,0.871]**$^{\downarrow}$	1.059 [0.993,1.103]$^{\sim}$	m^*=32 (MMMWW)	m^*=16 (MMWWW)
NIST–N_2	$P_0 = 1.0$ bar	50	1.103 [1.050,1.120]$^{\uparrow}$	1.103 [1.006,1.208]$^{\uparrow}$	1.002 [0.919,1.025]$^{\sim}$	– (MMMML)	– (MMMML)
CRAFTED	$P_0 = 0.01$ bar	1	1.195 [1.180,1.197]$^{\uparrow}$	1.072 [1.066,1.083]$^{\uparrow}$	1.111 [1.100,1.150]$^{\uparrow}$	– (LWMLL)	– (LWMLL)
CRAFTED	$P_0 = 0.01$ bar	50	1.146 [1.130,1.215]$^{\uparrow}$	1.039 [1.016,1.040]$^{\uparrow}$	1.145 [1.079,1.147]$^{\uparrow}$	– (LWMLL)	– (LWWML)
CRAFTED	$P_0 = 1.0$ bar	1	**0.609 [0.573,0.638]**$^{\downarrow}$	**0.560 [0.545,0.591]**$^{\downarrow}$	1.149 [1.036,1.214]$^{\uparrow}$	m^*=8 (LWWWW)	m^*=8 (LWWWW)
CRAFTED	$P_0 = 1.0$ bar	50	**0.724 [0.672,0.745]**$^{\downarrow}$	**0.609 [0.607,0.629]**$^{\downarrow}$	1.110 [1.105,1.129]$^{\uparrow}$	m^*=16 (LMWWW)	m^*=16 (LMWWW)

on NIST--N_2 at $k = 1$ (0.942× and 0.867×). However, when seeded in the dilute Henry regime ($P_0 = 0.01$ bar) on CRAFTED, transporting brackets across a 100× pressure jump (0.01 → 1.0 bar) yields wide rescaled intervals ($c_1, c_2 \sim 10^2$) while charging a slow initial seed solve, causing 64-query paid losses against WarmOnDemand (1.146×–1.195× for Transport; 1.039×–1.072× for Roots--Only). Similarly, on the 353-pair NIST--N_2 pool at $k = 50$, returning 14.2% of the pool leaves too few prunable candidates to amortize seed acquisition (1.099×–1.103×). Notably, when the seed anchor state is already warm in memory (warm-coverage evaluation in Fig. 4c), query execution is faster than WarmOnDemand from the very first query (0.33×–0.66× on CRAFTED).

D. Scaling Phase Space, Structural Descriptors, and Arrival Boundaries

Paid speedups scale monotonically with candidate pool size N and tighten when $k \ll N$. Fig. 4a maps the 32-query paid ratio (Transport / Incumbent) across the 4×4 (N, k) grid on CRAFTED. In both near and far regimes, every configuration with $N \geq 4{,}096$ achieves an all-block paid win across all ranking depths $k \in \{1, 10, 50, 200\}$ (0.57×–0.69× on near; 0.75×–0.96× on far). At $N = 1{,}024$, Transport wins all depths $k \leq 50$ (0.64×–0.76× on near; 0.82×–0.93× on far) and crosses 1.0× only at $k = 200$ (1.04× near, 1.19× far) where 19.5% of the pool is returned. At the smallest pool size $N = 256$, selective queries ($k \leq 10$) still win (0.72× near; 0.82×–0.84× far), whereas $k \in \{50, 200\}$ (19.5%–78.1% of the pool) incur paid losses (1.11×–1.65×).

Candidate pruning efficiency reflects target support and rank-k boundary gaps, and remains effective under small random arrivals while degrading gracefully under wide random jumps. Tables III and V connect these empirical phase boundaries to workload structure and stochastic query arrivals. Table III reports the eligible candidate support fraction (85.4%–87.5% on NIST--N_2 and 93.2%–96.1% on CRAFTED) alongside the normalized rank-k boundary gap $(s_k - s_{k+1})/\max(1, |s_k|)$ at the first and 64th queries. On CRAFTED, high support eligibility (93.2%–96.1%) means static support-cap filtering leaves nearly the entire pool active, yet transported brackets resolve even tight rank-k boundary gaps (4.6×10^{-6} to 4.0×10^{-3}) while eliminating 78.3%–99.4% of query-phase $\Pi_i(p)$ probes. Finally, Table V evaluates 16-query streams under independent normal random-arrival seeds (S1–S3) with *small* and *wide* step variances. On CRAFTED, Transport achieves all-block paid wins across all six *small*-arrival cells (0.689×–0.723× vs. LeanCap; 0.697×–0.759× vs. Incumbent) and wins on *wide* arrivals at $k = 1$ (0.840×–0.955× vs. LeanCap), whereas *wide* arrivals at $k = 50$ over only 16 queries become mixed (0.966×–0.991× vs. LeanCap; 1.000×–1.024× vs. Incumbent) and Roots--Only is 2%–8% faster (1.019×–1.084× for T/R). On the smaller NIST--N_2 pool ($N = 353$), Transport wins all three *small* $k = 1$ seeds against Incumbent (0.802×–0.895×), while *wide* $k = 50$ seeds do not amortize seed acquisition within 16 queries (1.154×–1.215× vs. Incumbent). Together, these diagnostics confirm that conditional root-enclosure transport excels over large candidate repositories ($N \geq 10^3$), selective top-k cuts ($k/N \leq 5\%$), and moderate or anchored continuation sweeps.

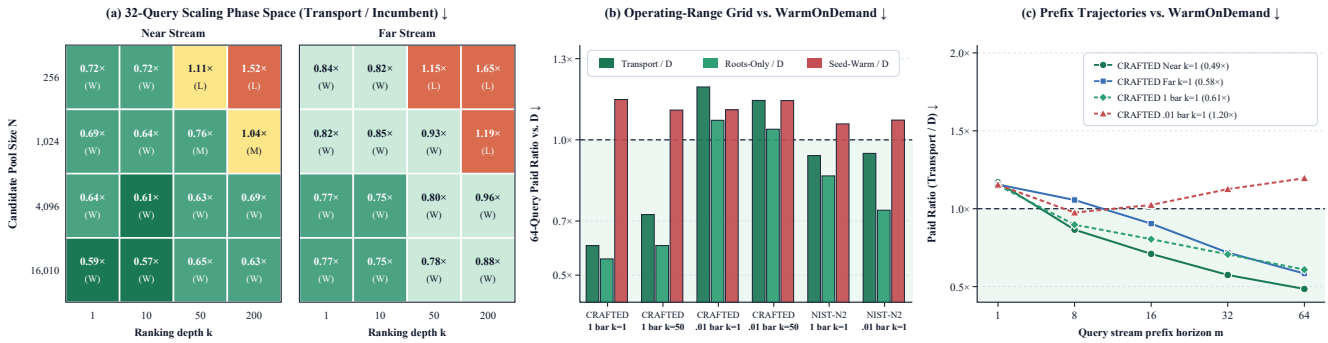


Fig. 4. **Root-enclosure transport achieves all-block paid speedups across large pools ($N \geq 4{,}096$) and 1 bar flue-gas sweeps, whereas small pools with large return fractions ($k/N \gtrsim 14\%$) and 0.01 bar dilute cold starts favor unseeded execution. (a)** 32-query paid lifecycle ratio heatmaps (`Transport / Incumbent`) across pool sizes $N \in \{256, 1{,}024, 4{,}096, 16{,}010\}$ and ranking depths $k \in \{1, 10, 50, 200\}$ on `CRAFTED` Near and Far streams; teal cells ($< 1.0\times$) denote paid speedups and warm amber/coral cells ($> 1.0\times$) denote small-pool, large-k losses. **(b)** Literature operating-range grid (0.01 vs. 1 bar) showing 64-query paid ratios and prefix trajectories against seed-free `WarmOnDemand (D)`. **(c)** Prefix ratio trajectories across horizons $m \in \{1, 8, 16, 32, 64\}$ against `WarmOnDemand` on `CRAFTED` and `NIST--N2`, alongside already-warm pool execution.

TABLE V

ROOT-ENCLOSURE TRANSPORT MAINTAINS ALL-BLOCK 16-QUERY PAID SPEEDUPS ON `CRAFTED` UNDER SMALL STOCHASTIC QUERY JUMPS ($0.689\times$–$0.759\times$) AND SELECTIVE WIDE JUMPS ($k = 1$), WHILE DEGRADING GRACEFULLY TOWARD PARITY AT $k = 50$ UNDER WIDE JUMPS. MEDIAN [min, max] PAID RATIOS (TRANSPORT/CONTROL) ACROSS THREE BLOCKS ON `NIST--N2` (LEFT) AND `CRAFTED` (RIGHT) OVER NORMAL SEEDS (S1–S3); TEAL ↓ = 3-BLOCK WIN, AMBER ∼ = MIXED, COPPER ↑ = 3-BLOCK LOSS.

			`NIST--N2` ($N = 353$ **pairs, 16 queries**)			**`CRAFTED`** ($N = 16{,}010$ **pairs, 16 queries**)		
Jump Regime	k	Seed	vs. `LeanCap`	vs. `Incumbent`	vs. `Roots--Only`	vs. `LeanCap`	vs. `Incumbent`	vs. `Roots--Only`
Small Jumps	1	S1	0.953 [0.781,1.030]∼	**0.802 [0.601,0.943]↓**	**0.958 [0.947,0.994]↓**	**0.720 [0.718,0.737]↓**	**0.759 [0.748,0.797]↓**	0.965 [0.962,1.036]∼
Small Jumps	1	S2	0.962 [0.949,1.015]∼	**0.853 [0.843,0.917]↓**	0.985 [0.956,1.036]∼	**0.723 [0.700,0.757]↓**	**0.758 [0.753,0.771]↓**	1.005 [1.001,1.046]↑
Small Jumps	1	S3	0.920 [0.826,1.002]∼	**0.895 [0.876,0.960]↓**	**0.969 [0.795,0.988]↓**	**0.690 [0.680,0.740]↓**	**0.738 [0.715,0.748]↓**	1.033 [0.967,1.097]∼
Small Jumps	50	S1	1.066 [0.955,1.200]∼	1.098 [0.928,1.283]∼	1.299 [0.940,1.414]∼	**0.707 [0.694,0.741]↓**	**0.753 [0.722,0.770]↓**	0.994 [0.990,1.035]∼
Small Jumps	50	S2	**0.882 [0.839,0.889]↓**	**0.979 [0.964,0.989]↓**	**0.947 [0.943,0.974]↓**	**0.700 [0.695,0.701]↓**	**0.743 [0.724,0.762]↓**	**0.989 [0.941,0.992]↓**
Small Jumps	50	S3	1.017 [0.910,1.199]∼	1.168 [1.023,1.168]↑	1.118 [0.985,1.298]∼	**0.689 [0.644,0.700]↓**	**0.697 [0.688,0.710]↓**	**0.923 [0.877,0.989]↓**
Wide Jumps	1	S1	0.963 [0.843,1.005]∼	**0.944 [0.927,0.996]↓**	1.020 [0.978,1.107]∼	**0.840 [0.809,0.984]↓**	**0.908 [0.811,0.946]↓**	1.075 [1.007,1.090]↑
Wide Jumps	1	S2	1.109 [0.849,1.150]∼	0.956 [0.762,1.103]∼	1.000 [0.895,1.197]∼	**0.955 [0.946,0.968]↓**	1.001 [0.998,1.012]∼	1.042 [0.962,1.062]∼
Wide Jumps	1	S3	**0.982 [0.950,1.000]↓**	**0.874 [0.866,0.932]↓**	**0.938 [0.907,0.980]↓**	**0.898 [0.892,0.901]↓**	**0.923 [0.913,0.969]↓**	1.019 [1.014,1.050]↑
Wide Jumps	50	S1	1.127 [1.084,1.239]↑	1.196 [0.977,1.376]∼	1.052 [1.029,1.107]↑	**0.966 [0.948,0.969]↓**	1.016 [0.973,1.040]∼	1.038 [1.022,1.048]↑
Wide Jumps	50	S2	1.068 [1.065,1.149]↑	1.215 [1.174,1.220]↑	1.059 [1.020,1.064]↑	0.991 [0.976,1.001]∼	1.024 [0.999,1.035]∼	1.076 [1.068,1.130]↑
Wide Jumps	50	S3	**0.925 [0.762,0.935]↓**	1.154 [1.151,1.207]↑	0.953 [0.928,1.022]∼	0.968 [0.944,1.006]∼	1.000 [0.960,1.029]∼	1.084 [1.081,1.137]↑

VIII. CONCLUSION AND OPERATING BOUNDARIES

We introduced **Conditional Root-Enclosure Transport**, an exact query-processing mechanism for repeated top-k ranking over implicit equilibrium scores in piecewise-linear adsorption isotherm repositories. Exploiting opposite-monotone partial-pressure rescaling (Theorem 1), our method transports prior solved root brackets and one-sided negative threshold witnesses into certified target-query root enclosures in $O(1)$ arithmetic per anchor, elevating historical root-finding state from a post-filter solver warm start into a pre-probe candidate-pruning primitive. Coupled with closed-form selectivity bounds (Proposition 1), outward-rounded interval residual testing against a guarded tolerance shell τ_-, and a static r_{i2}-ordered support-cap index, transported bounds contract mean target root intervals by $24.6\times$–$32.8\times$ on near continuation streams, drive 99.9% of upper-bound exclusions on `CRAFTED`, and cut 64-query paid cold-pool runtime by up to 54% while preserving exact output equivalence across all 11 policies.

Our lifecycle and boundary studies establish clear deployment rules for scientific database engines. First, because cold-pool seeding invests one full-pool pass at q_0, unseeded execution (`Incumbent` or `WarmOnDemand`) is preferable for single-shot queries ($m = 1$), small pools with large return fractions ($k/N \gtrsim 10\%$, e.g., $N = 353, k = 50$ or $N \leq 1{,}024, k = 200$), and dilute cold starts ($P_0 = 0.01$ bar) before $100\times$ pressure jumps. Second, once a seed pass is amortized ($m \geq 8$–16), two-slot positive-root retention (`Roots--Only`) offers a fast, low-overhead default (0%–8% faster than full witness retention in most cold-pool cells), whereas full two-slot witness transport (`Transport`) minimizes query-phase spreading-pressure probes under high-k sweeps (199.1K vs. 297.7K probes on `CRAFTED` far $k = 1$). Extending directional bracket transport to multicomponent equilibria and learned hybrid isotherm models is a promising avenue for future work.

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